Compiled by Maximus Pung Jun
H2 Mathematics (9758)
2026

2026 H2 Math Vectors I Homework 3: Solutions

Vector product: properties, parallel vectors, areas, collinearity. Official worked solutions for the sourced past-paper questions. Q1 True/False answers from the Basics Worksheet. Total: 46 marks (Q1 unmarked).

1 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\mathbf{a}\times\mathbf{a}=\mathbf{0}$. $\lvert\mathbf{a}\times\mathbf{a}\rvert=\lvert\mathbf{a}\rvert^2\sin 0^\circ=0$	**T**
(b)	$\mathbf{a}\times\mathbf{a}=0$. $\mathbf{a}\times\mathbf{a}=\mathbf{0}\neq 0$ (left side is the zero vector)	**F**
(c)	$\mathbf{i}\times\mathbf{j}=\mathbf{k}$. $\begin{pmatrix}1\\0\\0\end{pmatrix}\times\begin{pmatrix}0\\1\\0\end{pmatrix}=\begin{pmatrix}0\\0\\1\end{pmatrix}$	**T**
(d)	$\mathbf{a}\times\mathbf{b}=\mathbf{b}\times\mathbf{a}$. $\mathbf{i}\times\mathbf{j}=\mathbf{k},\ \mathbf{j}\times\mathbf{i}=-\mathbf{k}$	**F**
(e)	$\mathbf{a}\times\mathbf{b}=-(\mathbf{b}\times\mathbf{a})$. $\mathbf{a}\times\mathbf{b}=-\mathbf{b}\times\mathbf{a}$	**T**
(f)	$\mathbf{a}\times\mathbf{b}$ is a scalar. $\mathbf{i}\times\mathbf{j}=\mathbf{k}$ (a vector)	**F**
(g)	$\mathbf{k}\times\mathbf{j}=\mathbf{i}$. $\mathbf{j}\times\mathbf{k}=\mathbf{i}\ \therefore\ \mathbf{k}\times\mathbf{j}=-\mathbf{i}$	**F**

	Statement	**T/F**
(h)	$\begin{pmatrix}1\\2\\3\end{pmatrix}\times\begin{pmatrix}2\\4\\6\end{pmatrix}=\mathbf{0}$. $\begin{pmatrix}2\\4\\6\end{pmatrix}=2\begin{pmatrix}1\\2\\3\end{pmatrix}$	**T**
(i)	$\mathbf{a}\times(\mathbf{b}+\mathbf{c})=\mathbf{a}\times\mathbf{b}+\mathbf{a}\times\mathbf{c}$. Distributive law	**T**
(j)	$(\lambda\mathbf{a})\times\mathbf{b}=\lambda(\mathbf{a}\times\mathbf{b})$ for all $\lambda\in\mathbb{R}$. $(\lambda\mathbf{a})\times\mathbf{b}=\mathbf{a}\times(\lambda\mathbf{b})=\lambda(\mathbf{a}\times\mathbf{b})$	**T**
(k)	For non-zero and non-parallel **a** and **b**, $\mathbf{a}\times\mathbf{b}$ is perpendicular to both **a** and **b**. $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{a}=0,\ (\mathbf{a}\times\mathbf{b})\cdot\mathbf{b}=0$	**T**
(l)	$(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}$ is a scalar. $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}=\text{vector}\cdot\text{vector}$	**T**
(m)	$\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta$, where θ is the angle between **a** and **b**. Definition of vector product	**T**
(n)	$\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$. $\lvert\mathbf{i}\times\mathbf{i}\rvert=0\neq1$	**F**
(o)	If $\mathbf{a}\times\mathbf{b}=\mathbf{0}$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$. $\mathbf{i}\times2\mathbf{i}=\mathbf{0},\ \mathbf{i}\neq\mathbf{0},\ 2\mathbf{i}\neq\mathbf{0}$	**F**
(p)	For non-zero **a** and **b**, $\mathbf{a}\times\mathbf{b}=\mathbf{0}$ means that **a** is parallel to **b**. $\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta=0\ \therefore\ \theta=0^\circ$ or 180°	**T**
(q)	The area of the parallelogram with adjacent sides **a** and **b** is $\lvert\mathbf{a}\times\mathbf{b}\rvert$. Area $=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta=\lvert\mathbf{a}\times\mathbf{b}\rvert$	**T**
(r)	The area of triangle OAB is $\lvert\overrightarrow{OA}\times\overrightarrow{OB}\rvert$. Area $=\frac{1}{2}\lvert\overrightarrow{OA}\times\overrightarrow{OB}\rvert$	**F**

	Statement	**T/F**
(s)	$\mathbf{a}\times(\mathbf{b}\cdot\mathbf{c})$ is a vector. $\mathbf{b}\cdot\mathbf{c}$ is a scalar $\therefore$ meaningless	**F**
(t)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, if $\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$, then $\mathbf{a}$ is perpendicular to $\mathbf{b}$. $\sin\theta=1 \therefore \theta=90^\circ$	**T**
(u)	$(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}=\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$. both $=\det(\mathbf{a}\ \mathbf{b}\ \mathbf{c})$	**T**
(v)	$\lvert\mathbf{a}\times\mathbf{b}\rvert^2+(\mathbf{a}\cdot\mathbf{b})^2=\lvert\mathbf{a}\rvert^2\lvert\mathbf{b}\rvert^2$. $\lvert\mathbf{a}\rvert^2\lvert\mathbf{b}\rvert^2(\sin^2\theta+\cos^2\theta)=\lvert\mathbf{a}\rvert^2\lvert\mathbf{b}\rvert^2$	**T**
(w)	$\mathbf{a}\times\mathbf{b}=\mathbf{a}\times\mathbf{c}$ means that $\mathbf{b}=\mathbf{c}$. $\mathbf{i}\times\mathbf{j}=\mathbf{k}=\mathbf{i}\times(\mathbf{i}+\mathbf{j}),\ \mathbf{j}\neq\mathbf{i}+\mathbf{j}$	**F**
(x)	$\mathbf{a}\times(\mathbf{b}\times\mathbf{c})=(\mathbf{a}\times\mathbf{b})\times\mathbf{c}$. $\mathbf{i}\times(\mathbf{i}\times\mathbf{j})=\mathbf{i}\times\mathbf{k}=-\mathbf{j},\ (\mathbf{i}\times\mathbf{i})\times\mathbf{j}=\mathbf{0}$	**F**
(y)	The shortest distance from B to the line through A parallel to $\mathbf{d}$ is $\lvert\overrightarrow{AB}\times\mathbf{d}\rvert$. distance $=\lvert\overrightarrow{AB}\times\hat{\mathbf{d}}\rvert=\dfrac{\lvert\overrightarrow{AB}\times\mathbf{d}\rvert}{\lvert\mathbf{d}\rvert}$	**F**
(z)	If $\mathbf{a}\times\mathbf{b}=\mathbf{0}$ and $\mathbf{a}\cdot\mathbf{b}=0$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$. $\lvert\mathbf{a}\rvert^2\lvert\mathbf{b}\rvert^2=\lvert\mathbf{a}\times\mathbf{b}\rvert^2+(\mathbf{a}\cdot\mathbf{b})^2=0$	**T**

2 [JPJC 2024 Promo Q11]

Referred to the origin O, points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are non-zero and non-parallel vectors. Point C is the midpoint of OA and point D lies on BC such that $BD:DC=2:3$.

(i) Find the position vectors $\overrightarrow{OC}$ and $\overrightarrow{OD}$, giving your answers in terms of $\mathbf{a}$ and $\mathbf{b}$. [2]

(ii) Find the area of triangle OCD in the form of $k|\mathbf{a}\times\mathbf{b}|$, where k is a constant to be found. [3]

(iii) Given that $|\mathbf{a}|=2$, $|\mathbf{b}|=\sqrt{2}$ and angle AOB is $\frac{\pi}{4}$ radian, using scalar product, show that CD is perpendicular to OA. [3]

(iv) Given instead that $\mathbf{a}$ is a unit vector, give a geometrical interpretation of $|\mathbf{a}\cdot\mathbf{b}|$. [1]

Solution		Comments
(i) **[2]**	Given $\overrightarrow{OA}=\mathbf{a}$, $\overrightarrow{OB}=\mathbf{b}$, $\overrightarrow{OC}=\frac{1}{2}\mathbf{a}$. $\overrightarrow{OD}=\frac{2\mathbf{c}+3\mathbf{b}}{5}=\frac{1}{5}\left[2\left(\frac{1}{2}\mathbf{a}\right)+3\mathbf{b}\right]=\frac{1}{5}(\mathbf{a}+3\mathbf{b})$.	
(ii) **[3]**	Area of triangle $OCD=\frac{1}{2}\lvert\overrightarrow{OC}\times\overrightarrow{OD}\rvert$ $=\frac{1}{2}\left\lvert\frac{1}{2}\mathbf{a}\times\frac{1}{5}(\mathbf{a}+3\mathbf{b})\right\rvert$ $=\frac{1}{2}\left\lvert\frac{1}{10}(\mathbf{a}\times\mathbf{a})+\frac{3}{10}(\mathbf{a}\times\mathbf{b})\right\rvert$ $=\frac{1}{2}\left\lvert\frac{3}{10}(\mathbf{a}\times\mathbf{b})\right\rvert \qquad (\mathbf{a}\times\mathbf{a}=\mathbf{0})$ $=\frac{3}{20}\lvert\mathbf{a}\times\mathbf{b}\rvert, \qquad k=\frac{3}{20}$.	
(iii) **[3]**	$\overrightarrow{CD}=\overrightarrow{OD}-\overrightarrow{OC}=\frac{1}{5}(\mathbf{a}+3\mathbf{b})-\frac{1}{2}\mathbf{a}=\frac{3}{5}\mathbf{b}-\frac{3}{10}\mathbf{a}$ $\overrightarrow{CD}\cdot\overrightarrow{OA}=\left(\frac{3}{5}\mathbf{b}-\frac{3}{10}\mathbf{a}\right)\cdot\mathbf{a}$ $=\frac{3}{5}\mathbf{b}\cdot\mathbf{a}-\frac{3}{10}\mathbf{a}\cdot\mathbf{a}$ $=\frac{3}{5}\lvert\mathbf{b}\rvert\lvert\mathbf{a}\rvert\cos\frac{\pi}{4}-\frac{3}{10}\lvert\mathbf{a}\rvert^2$ $=\frac{3}{5}(\sqrt{2})(2)\frac{\sqrt{2}}{2}-\frac{3}{10}(2)^2=\frac{6}{5}-\frac{12}{10}=0.$ Since $\overrightarrow{CD}\cdot\overrightarrow{OA}=0$, CD is perpendicular to OA. (Shown)	
(iv) **[1]**	$\lvert\mathbf{a}\cdot\mathbf{b}\rvert$ is the length of projection of $\overrightarrow{OB}$ onto $\overrightarrow{OA}$ (or a line or vector parallel to $\overrightarrow{OA}$).	

3 [MI 2024 Promo Q4]

Referred to the origin O, the points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$. C lies on AB such that $AC:AB=1:3$.

(i) Find $\mathbf{c}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [2]

(ii) Show that the area of triangle OAC is given by $k|\mathbf{a}\times\mathbf{b}|$, where k is a constant to be determined. [2]

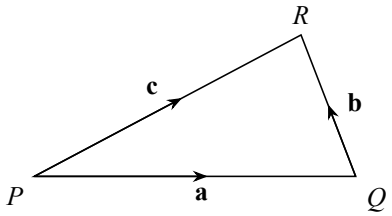
It is now given that $\mathbf{a}$ and $\mathbf{b}$ are unit vectors.

(iii) Using the result in part (i), find the value of $(2\mathbf{a}-\mathbf{b})\cdot\mathbf{c}$. [3]

Solution		**Comments**
(i) **[2]**	$AC:AB=1:3 \quad\Rightarrow\quad AC:CB=1:2$ $\mathbf{c}=\dfrac{(1)\mathbf{b}+(2)\mathbf{a}}{1+2}=\dfrac{2\mathbf{a}+\mathbf{b}}{3}.$	
(ii) **[2]**	Area of triangle $OAC=\dfrac{1}{3}$ Area of triangle OAB $=\dfrac{1}{3}\times\dfrac{1}{2}\lvert\overrightarrow{OA}\times\overrightarrow{OB}\rvert$ $=\dfrac{1}{6}\lvert\mathbf{a}\times\mathbf{b}\rvert, \qquad k=\dfrac{1}{6}.$	
(iii) **[3]**	$(2\mathbf{a}-\mathbf{b})\cdot\mathbf{c}=(2\mathbf{a}-\mathbf{b})\cdot\left(\dfrac{2\mathbf{a}+\mathbf{b}}{3}\right)$ $=\dfrac{1}{3}(2\mathbf{a}-\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b})$ $=\dfrac{1}{3}(4\mathbf{a}\cdot\mathbf{a}+2\mathbf{a}\cdot\mathbf{b}-2\mathbf{b}\cdot\mathbf{a}-\mathbf{b}\cdot\mathbf{b})$ $=\dfrac{1}{3}(4\lvert\mathbf{a}\rvert^2-\lvert\mathbf{b}\rvert^2)$ $=\dfrac{1}{3}(4(1)^2-(1)^2)$ since $\mathbf{a}$, $\mathbf{b}$ are unit vectors $=1.$	

4 [ASRJC 2024 Promo Q5]

In triangle PQR, the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ represent the vectors $\overrightarrow{PQ}$, $\overrightarrow{QR}$ and $\overrightarrow{PR}$ respectively.



(a) State a geometrical interpretation of $|\mathbf{c}\times\hat{\mathbf{a}}|$ and hence show that the area of the triangle PQR can be written as $\frac{1}{2}|\mathbf{c}\times\mathbf{a}|$ units^2. [3]

(b) With clear explanations, show that $|\mathbf{c}\times\mathbf{a}|=|\mathbf{b}\times\mathbf{a}|$ and deduce that

$$\frac{|\mathbf{c}|}{\sin\angle PQR}=\frac{|\mathbf{b}|}{\sin\angle QPR} \quad \text{(Sine Rule).}$$

[2]

Solution		**Comments**
(a) **[3]**	$\|\mathbf{c}\times\hat{\mathbf{a}}\|$ represents the perpendicular distance from the point R to the line PQ. Area of $\triangle PQR=\frac{1}{2}\times\text{base}\times\text{height}$ $=\frac{1}{2}\|\mathbf{a}\|\,\|\mathbf{c}\times\hat{\mathbf{a}}\|$ $=\frac{1}{2}\|\mathbf{a}\|\left\|\mathbf{c}\times\frac{\mathbf{a}}{\|\mathbf{a}\|}\right\|$ $=\frac{1}{2}\|\mathbf{c}\times\mathbf{a}\|$ units^2. (Shown)	
(b) **[2]**	By replacing $\mathbf{a}$ with $-\mathbf{a}$ and $\mathbf{c}$ with $\mathbf{b}$ in part (a), Area of $\triangle PQR=\frac{1}{2}\|\mathbf{b}\times(-\mathbf{a})\|=\frac{1}{2}\|\mathbf{b}\times\mathbf{a}\|$ $\frac{1}{2}\|\mathbf{c}\times\mathbf{a}\|=\frac{1}{2}\|\mathbf{b}\times\mathbf{a}\|$ $\|\mathbf{c}\times\mathbf{a}\|=\|\mathbf{b}\times\mathbf{a}\|$ (Shown) $\|\mathbf{c}\|\|\mathbf{a}\|\sin\angle QPR=\|\mathbf{b}\|\|\mathbf{a}\|\sin\angle PQR$ $\frac{\|\mathbf{c}\|}{\sin\angle PQR}=\frac{\|\mathbf{b}\|}{\sin\angle QPR}$. (Shown)	

5 [CJC 2024 Promo Q7]

(a) Three non-zero vectors **p**, **q** and **r** are such that $3\mathbf{p}\times\mathbf{q}=\mathbf{r}\times\mathbf{p}$.

Find a linear relationship between **p**, **q** and **r**. [3]

(b) Referred to the origin O, the points A and B have position vectors **a** and **b**.

It is given that **a** and **b** are non-parallel unit vectors.

The point B divides AC in the ratio $1:3$.

The point D lies on OB produced such that $OB:OD=1:m$, where $m\in\mathbb{Z}$, $m\geq 2$.

Given that $\overrightarrow{AB}$ is perpendicular to $\overrightarrow{CD}$, find the numerical value of m. [4]

Solution		Comments
(a) **[3]**	$3\mathbf{p}\times\mathbf{q}=\mathbf{r}\times\mathbf{p}$ $(3\mathbf{p}\times\mathbf{q})-(\mathbf{r}\times\mathbf{p})=\mathbf{0}$ $(3\mathbf{p}\times\mathbf{q})+(\mathbf{p}\times\mathbf{r})=\mathbf{0}$ $\mathbf{p}\times(3\mathbf{q}+\mathbf{r})=\mathbf{0}$ $\mathbf{p}\parallel 3\mathbf{q}+\mathbf{r}$ $\mathbf{p}=k(3\mathbf{q}+\mathbf{r}),\quad k\in\mathbb{R}\setminus\{0\}.$ *Verification note (not part of the official scheme):* the printed last line assumes $3\mathbf{q}+\mathbf{r}\neq\mathbf{0}$. The question also permits $3\mathbf{q}+\mathbf{r}=\mathbf{0}$. The unrestricted relationship is $3\mathbf{q}+\mathbf{r}=\lambda\mathbf{p}$, $\lambda\in\mathbb{R}$, including $\lambda=0$.	
(b) **[4]**	$\overrightarrow{OB}=\dfrac{3\overrightarrow{OA}+\overrightarrow{OC}}{4}$ $\overrightarrow{OC}=4\mathbf{b}-3\mathbf{a}$ $\overrightarrow{OD}=m\overrightarrow{OB}=m\mathbf{b}$ $\overrightarrow{AB}=\mathbf{b}-\mathbf{a}$ $\overrightarrow{CD}=m\mathbf{b}-(4\mathbf{b}-3\mathbf{a})=3\mathbf{a}+(m-4)\mathbf{b}.$ Given that $\overrightarrow{AB}$ is perpendicular to $\overrightarrow{CD}$, $\overrightarrow{AB}\cdot\overrightarrow{CD}=0$ $(\mathbf{b}-\mathbf{a})\cdot[3\mathbf{a}+(m-4)\mathbf{b}]=0$ $3\mathbf{a}\cdot\mathbf{b}+(m-4)\mathbf{b}\cdot\mathbf{b}-3\mathbf{a}\cdot\mathbf{a}-(m-4)\mathbf{a}\cdot\mathbf{b}=0$ $3\mathbf{a}\cdot\mathbf{b}+(m-4)\lvert\mathbf{b}\rvert^2-3\lvert\mathbf{a}\rvert^2-(m-4)\mathbf{a}\cdot\mathbf{b}=0$ $(m-4)-3-(m-7)\mathbf{a}\cdot\mathbf{b}=0$ $m(1-\mathbf{a}\cdot\mathbf{b})=3-7\mathbf{a}\cdot\mathbf{b}+4$ $m=\dfrac{3-7\mathbf{a}\cdot\mathbf{b}+4}{1-\mathbf{a}\cdot\mathbf{b}}$ $=\dfrac{7(1-\mathbf{a}\cdot\mathbf{b})}{1-\mathbf{a}\cdot\mathbf{b}}=7.$	

6 [SAJC 2024 Promo Q9]

(a) Relative to the origin O, the points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are non-parallel. It is given that $|\mathbf{a}|=9$, $|\mathbf{b}|=1$ and $|2\mathbf{a}+\mathbf{b}|=2\sqrt{74}$.

(i) By considering $(2\mathbf{a}+\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b})$, show that $\mathbf{a}\cdot\mathbf{b}=-\frac{29}{4}$. [3]

(ii) Give a geometrical meaning of $|\mathbf{a}\cdot\mathbf{b}|$. [1]

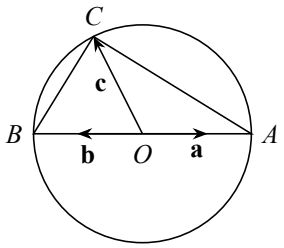
(iii) The points P, Q and R have position vectors $7\mathbf{a}-5\mathbf{b}$, $6\mathbf{a}+5\mathbf{b}$ and $9\mathbf{a}+\lambda\mathbf{b}$ respectively. Given that these three points are collinear, find the value of λ. [3]

(b) In the triangle OUV where O is the origin, the position vectors of the points U and V are $\mathbf{u}$ and $\mathbf{v}$ respectively. The point W is the midpoint of OU, and the point X has position vector given by $\frac{1}{16}\mathbf{u}+\frac{3}{4}\mathbf{v}$. Show that the area of triangle UWX can be written as $m|\mathbf{u}\times\mathbf{v}|$ where m is a constant to be found. [3]

Solution		Comments
(a)(i) **[3]**	$(2\mathbf{a}+\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b}) = 4(\mathbf{a}\cdot\mathbf{a})+2(\mathbf{b}\cdot\mathbf{a})+2(\mathbf{a}\cdot\mathbf{b})+(\mathbf{b}\cdot\mathbf{b})$ $= 4\lvert\mathbf{a}\rvert^2+4(\mathbf{a}\cdot\mathbf{b})+\lvert\mathbf{b}\rvert^2$ $\lvert2\mathbf{a}+\mathbf{b}\rvert^2 = 4\lvert\mathbf{a}\rvert^2+4(\mathbf{a}\cdot\mathbf{b})+\lvert\mathbf{b}\rvert^2$ $(2\sqrt{74})^2 = 4(9^2)+4(\mathbf{a}\cdot\mathbf{b})+(1^2)$ $296 = 4(9^2)+4(\mathbf{a}\cdot\mathbf{b})+(1^2)$ $4(\mathbf{a}\cdot\mathbf{b}) = 296-4(9^2)-(1^2)$ $\mathbf{a}\cdot\mathbf{b} = -\frac{29}{4}.$	
(a)(ii) **[1]**	$\lvert\mathbf{a}\cdot\mathbf{b}\rvert$ is the length of projection of $\overrightarrow{OA}$ onto $\overrightarrow{OB}$.	
(a)(iii) **[3]**	If P, Q and R are collinear, then $k\overrightarrow{PQ}=\overrightarrow{PR}$. $k[(6\mathbf{a}+5\mathbf{b})-(7\mathbf{a}-5\mathbf{b})] = [(9\mathbf{a}+\lambda\mathbf{b})-(7\mathbf{a}-5\mathbf{b})]$ $k[-\mathbf{a}+10\mathbf{b}] = [2\mathbf{a}+(\lambda+5)\mathbf{b}].$ Since **a** and **b** are non-parallel, non-zero vectors, $k=-2$ $\lambda+5=10k=-20$ $\lambda=-25.$	
(b) **[3]**	$\overrightarrow{UX}=\overrightarrow{OX}-\overrightarrow{OU}=\frac{1}{16}\mathbf{u}+\frac{3}{4}\mathbf{v}-\mathbf{u}=\frac{3}{4}\mathbf{v}-\frac{15}{16}\mathbf{u}$ $\overrightarrow{UW}=\overrightarrow{OW}-\overrightarrow{OU}=\frac{1}{2}\mathbf{u}-\mathbf{u}=-\frac{1}{2}\mathbf{u}$ Area of triangle $UWX=\frac{1}{2}\lvert\overrightarrow{UW}\times\overrightarrow{UX}\rvert$ $=\frac{1}{2}\left\lvert\frac{1}{2}\mathbf{u}\times\left(\frac{3}{4}\mathbf{v}-\frac{15}{16}\mathbf{u}\right)\right\rvert$ $=\frac{3}{16}\lvert\mathbf{u}\times\mathbf{v}\rvert \qquad (\mathbf{u}\times\mathbf{u}=\mathbf{0})$ $m=\frac{3}{16}.$	

7 [CJC 2025 Promo Q6]

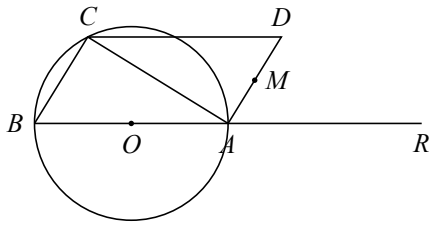
The diagram shows a unit circle with centre O and diameter AB. The point C lies on the circumference of the circle. The position vectors of A, B and C are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively.



(a) By considering an appropriate scalar product, show that AC and BC are perpendicular. [4]

(b) Give a geometrical interpretation of $|\mathbf{a}\cdot(\mathbf{c}-\mathbf{a})|$. [1]

The point D is such that $ABCD$ is a parallelogram. The point M is the mid-point of AD and the point R lies on OA produced such that $OA:AR=1:2$.



(c) By considering $\overrightarrow{CM}\times\overrightarrow{CR}$ or otherwise, show that the points C, M and R are collinear. [3]

Solution		Comments
(a) **[4]**	$\overrightarrow{AC} = \mathbf{c} - \mathbf{a}, \qquad \overrightarrow{BC} = \mathbf{c} - \mathbf{b}$ $\overrightarrow{AC} \cdot \overrightarrow{BC} = (\mathbf{c} - \mathbf{a}) \cdot (\mathbf{c} - \mathbf{b})$ $= (\mathbf{c} - \mathbf{a}) \cdot (\mathbf{c} - (-\mathbf{a}))$ $= (\mathbf{c} - \mathbf{a}) \cdot (\mathbf{c} + \mathbf{a})$ $= \mathbf{c} \cdot \mathbf{c} - \mathbf{a} \cdot \mathbf{a}$ $= \lvert\mathbf{c}\rvert^2 - \lvert\mathbf{a}\rvert^2$ $= 0 \qquad (\lvert\mathbf{c}\rvert = \lvert\mathbf{a}\rvert = \text{radius of circle}).$ Hence AC and BC are perpendicular.	
(b) **[1]**	$\lvert\mathbf{a} \cdot (\mathbf{c} - \mathbf{a})\rvert$ is the length of projection of $\overrightarrow{AC}$ onto $\overrightarrow{OA}$.	
(c) **[3]**	By Ratio Theorem, $\overrightarrow{OM} = \dfrac{\overrightarrow{OD} + \overrightarrow{OA}}{2}$ $= \dfrac{1}{2}(\mathbf{c} + 2\mathbf{a} + \mathbf{a}) = \dfrac{1}{2}\mathbf{c} + \dfrac{3}{2}\mathbf{a}$ $\overrightarrow{CM} \times \overrightarrow{CR} = (\overrightarrow{OM} - \overrightarrow{OC}) \times (\overrightarrow{OR} - \overrightarrow{OC})$ $= \left[\dfrac{1}{2}\mathbf{c} + \dfrac{3}{2}\mathbf{a} - \mathbf{c}\right] \times (3\mathbf{a} - \mathbf{c})$ $= \left(\dfrac{3}{2}\mathbf{a} - \dfrac{1}{2}\mathbf{c}\right) \times (3\mathbf{a} - \mathbf{c})$ $= \dfrac{9}{2}\mathbf{a} \times \mathbf{a} - \dfrac{3}{2}\mathbf{a} \times \mathbf{c} - \dfrac{3}{2}\mathbf{c} \times \mathbf{a} + \dfrac{1}{2}\mathbf{c} \times \mathbf{c}$ $= \mathbf{0} - \dfrac{3}{2}\mathbf{a} \times \mathbf{c} + \dfrac{3}{2}\mathbf{a} \times \mathbf{c} + \mathbf{0}$ $= \mathbf{0}.$ $\overrightarrow{CM}$ and $\overrightarrow{CR}$ are parallel. Since CR and CM are parallel and C is a common point, hence C, M and R are collinear.	

END OF SOLUTIONS